HARSH JALNEKAR

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SUMMARY

Backend Engineer with professional experience designing secure, scalable backend systems for multi-tenant SaaS applications using Java, Spring Boot, PostgreSQL, RabbitMQ, Keycloak, and Docker. Experienced in REST API development, event-driven architectures, tenant-aware authorization, distributed messaging, and AI-powered backend workflows using local LLMs. Passionate about solving complex engineering problems and building reliable, production-ready software.

EXPERIENCE

Backend Engineer | *NextGen Digi Lab, Pune, India* Dec 2024 – Present

- Designed and developed scalable RESTful APIs for a multi-tenant SaaS platform using Java and Spring Boot, enabling secure communication between backend services while maintaining clean service and repository abstractions.
- Modernized legacy database access by migrating to Spring Data JPA and Hibernate, improving maintainability, transaction management, and efficient PostgreSQL resource utilization.
- Designed tenant-aware relational database schemas and entity relationships to enforce secure data isolation and support scalable multi-tenant architecture.
- Designed and implemented tenant-aware authentication and authorization workflows using Keycloak, OAuth 2.0, and Role-Based Access Control (RBAC) to secure enterprise applications.
- Built event-driven backend workflows using RabbitMQ to decouple asynchronous processing and improve application responsiveness and reliability.
- Developed an on-premises document summarization pipeline using Ollama, implementing text extraction, semantic chunking, prompt engineering, and deterministic response validation while ensuring sensitive data remained within the organization's infrastructure.
- Implemented hybrid AES-RSA encryption workflows to secure sensitive business data and protect multi-tenant application data.
- Built AI-assisted internal development workflows to automate repetitive engineering tasks and accelerate feature prototyping.
- Supported deployment and operational workflows using Linux, Docker, Nginx, GitHub Actions, and Git across development and staging environments.

Java Trainee | *PRADO Pvt Ltd, Pune, India* Jan 2024 – Oct 2024

- Built backend systems for real-time hardware monitoring applications using core Java, implementing multithreaded synchronization logic for sensor arrays.
- Programmed network proxy server workflows to manage and restrict communication channels within secure laboratory environments.
- Refactored network sockets and stream handling to reduce serialization latency and increase monitoring data throughput.

EDUCATION

Vishwakarma Institute of Technology, *Pune, India* 2020 – 2024
Bachelor of Technology in Electronics & Telecommunication Engineering

PUBLICATIONS

Stock Market Prediction Using LSTM

International Journal for Research in Applied Science and Engineering Technology (IJRASET) | December 2022

- Co-authored a peer-reviewed research paper on applying **Long Short-Term Memory (LSTM)** neural networks for stock market trend prediction and financial time-series forecasting.
- Explored deep learning techniques to improve predictive accuracy by modelling sequential market data and identifying temporal patterns.
- **Publication:** Available on ResearchGate.

CERTIFICATIONS & ACADEMIC PROGRAMS

- Google Cloud Gen AI Academy – APAC Edition
- Vector Search with Alloy DB – Google Cloud
- Engineer AI Agents with Agent Development Kit (ADK) – Google Developer Program

LEADERSHIP & VOLUNTEERING

Advisory Board Member | *Antariksh Astronomy Club, Pune* 2022 – 2024

- Progressed from club member to Advisory Board Member through active involvement in astronomy outreach and educational initiatives.
- Coordinated public space-science outreach initiatives and educational events for student communities.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, SQL
- **Frameworks & Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate ORM, REST APIs
- **Databases & Messaging:** PostgreSQL, RabbitMQ, MQTT
- **System Design:** Microservices, Multi-Tenant Architecture, Event-Driven Systems, Domain-Driven Design (DDD), Relational Database Design
- **AI & Intelligent Systems:** Ollama, LLM Integration, Prompt Engineering, Semantic Chunking, AI Workflow Automation
- **Security:** OAuth 2.0, Keycloak, RBAC, AES, RSA, Hybrid Encryption
- **DevOps & Infrastructure:** Docker, Linux, Nginx, Git, GitHub Actions, Maven